

KRISHNA ANNAVARAM

Generative AI Engineer | Agentic AI, RAG & Enterprise Modernization

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PROFESSIONAL SUMMARY

Generative AI Engineer with 5+ years building production LLM systems, RAG and Graph-RAG pipelines, and multi-agent orchestration frameworks across enterprise modernization, financial risk, and healthcare. Currently architecting reusable AI-assisted engineering harnesses on Google Cloud for legacy reverse engineering, quantitative risk intelligence, code-complexity assessment, Spring Boot migration, and adaptive vulnerability remediation, each built on an evidence-first pattern of deterministic analysis, governed AI reasoning, full traceability, and human approval gates. Prior delivery includes a hospital-deployed Graph-RAG clinical decision support system at 91% retrieval accuracy with a 45% reduction in unsafe recommendations, 85% automation of patient intake, and a 50% cut in physician review time through fine-tuned transformers. Certified as a Microsoft Azure AI Engineer Associate (AI-102) and AWS Certified AI Practitioner, with depth across LangGraph, A2A/MCP, Claude, Azure OpenAI (GPT-4o), Neo4j, Qdrant/FAISS, OpenRewrite, ANTLR4, and zero-downtime MLOps on GCP, Azure, and AWS.

CORE TECHNICAL SKILLS

GenAI & Agents	Claude (Sonnet/Opus), Azure OpenAI (GPT-4o, GPT-4), LangGraph, LangChain, LangSmith, Agentic AI, Multi-Agent Orchestration, A2A & MCP Protocols, ReAct, Tool Use & Function Calling, RAG, Graph-RAG, Prompt Engineering, Chain-of-Thought, Few-Shot Learning, Context Engineering
Fine-Tuning & Evaluation	LoRA, QLoRA, PEFT, Adapter & Instruction Tuning, RLHF, Long-T5, BART, BERT, RoBERTa, LLaMA, Mistral, Hugging Face Transformers · LLM-as-Judge Evaluation, Hallucination & Groundedness Testing, Guardrails, Prompt Versioning, Token & Cost Optimization, Latency Benchmarking
Modernization & AppSec	ANTLR4, sqlglot, OpenRewrite, Code & Dependency Graphs, Static Analysis, CWE Classification, Root-Cause & Blast-Radius Analysis, Compiler-Guided Repair, Remediation Validation, Differential & Regression Testing, Red-Team Validation
Retrieval & Data	Neo4j Knowledge Graphs, Qdrant, FAISS, Pinecone, Chroma, Milvus, Weaviate, PostgreSQL/pgvector, Redis Semantic Cache, MongoDB, Cosmos DB, BigQuery · Hybrid Dense-Sparse Retrieval, Re-Ranking, Chunking Strategies, Sentence Transformers, FHIR-Based EHR Integration
Cloud & MLOps	GCP (Vertex AI, BigQuery, Cloud Run, Dataflow) · Azure (Azure OpenAI, Azure ML, AKS, Azure Speech, Functions, DevOps, Databricks) · AWS (SageMaker, Bedrock, Lambda, S3, Glue, CodePipeline) · Docker, Kubernetes, Terraform, GitHub Actions, Jenkins, Airflow, MLflow, Model Monitoring, Canary & Blue-Green Deployments
ML & Programming	Python (Expert), SQL, Java, Bash, TypeScript · PyTorch, TensorFlow, scikit-learn, Keras, XGBoost, LightGBM, NLTK, spaCy, Transformers, Named Entity Recognition, Text Classification · Pandas, NumPy, Polars, ETL Pipelines, FastAPI, Streamlit

PROFESSIONAL EXPERIENCE

Generative AI Engineer <i>Virtusa Corporation United States</i>	Jun 2026 – Present
- Engineered STATUTE, an 8-stage PL/SQL reverse-engineering harness that converted legacy Oracle code into 41 traceable business requirements, BRDs, ERDs, process flows, and gap registers backed by a 353-node / 769-relationship knowledge graph, validating the deterministic pipeline through 414 automated checks before Google Cloud release. - Designed Smart Quant, a 3-agent expert reasoning and bounded-negotiation platform that validates market-risk methodology, required inputs, available capabilities, and minimum necessary data access before executing VaR, Expected Shortfall, stress, sensitivity, and yield-curve analytics across 267K+ market observations and 56 governed capabilities. - Reduced Smart Quant's incomplete-request path from 5 A2A handoffs, 4 vector searches, and 4 model calls to 2 handoffs, 0 searches, and 1 model call by introducing deterministic preflight completeness checks, bounded negotiation, and capability-aware execution ahead of expensive retrieval and computation. - Built an Adaptive Legacy Code Complexity Harness with 20 language-agnostic analyzers across cyclomatic, cognitive, coupling, dependency, architecture, database, runtime, testability, and maintainability dimensions, enabling source-to-target migration risk and effort comparison via normalized code representations. - Orchestrated BootShift, a 20-stage Spring Boot modernization harness integrating repository discovery, dependency and application graphs, target resolution, OpenRewrite transformations, compiler-guided repair, runtime and differential validation, and migration provenance; enforced 31 safety controls and validated 190 automated tests. - Extended a 7-agent, 9-stage CWE vulnerability detection and adaptive auto-remediation harness with root-cause analysis, CWE classification, blast-radius assessment, controlled patching, and red-team plus regression validation, adding similarity-driven research that derives fixes from five analogous security cases when no vetted patch exists.	

- Led POC delivery end to end — architecture, implementation, cross-project CI/CD, automated quality gates, technical reviews, documentation, Google Cloud releases, and stakeholder and management demos — translating enterprise GenAI, modernization, and security requirements into reusable engineering platforms.

Generative AI Engineer
Ideate Technologies LLC | United States

Mar 2026 – Jun 2026

- Led end-to-end delivery of a Graph-RAG Clinical Decision Support System for U.S. hospital clients, unifying Neo4j knowledge graphs, FAISS vector search, and Azure OpenAI GPT-4o into a real-time diagnostic reasoning pipeline that reached 91% retrieval accuracy and cut unsafe clinical recommendations by 45%.
- Directed a voice-based patient intake platform using Azure Speech-to-Text, LangGraph multi-agent orchestration, Azure OpenAI, and Cosmos DB, automating 85% of clinical data capture and reducing intake cycle time by 60% across live hospital workflows.
- Architected stateful A2A/MCP agent workflows with live FHIR-based EHR integration and fine-tuned Long-T5 and BART models using PEFT/LoRA on clinical corpora, reducing physician review time by 50% while delivering nurse-to-doctor handoff summaries at 92% completeness.
- Established zero-downtime GenAI MLOps on AKS, Azure ML, MLflow, Terraform, and GitHub Actions, cutting deployment effort by 70%, accelerating release cycles by 40%, and sustaining 99.9% uptime across production clinical services.

Graduate Teaching Assistant — Generative AI
University of North Texas | Denton, TX

Aug 2024 – May 2025

- Built and deployed an AI research assistant on FAISS, PostgreSQL, and Redis that delivered 88–92% retrieval relevance across thousands of student and faculty graduate-program queries, replacing manual literature search.
- Developed a full-stack RAG pipeline on Azure with LangGraph, Sentence Transformer embeddings, and hybrid dense-sparse retrieval across 10,000+ academic documents, holding sub-second responses under concurrent multi-user load.
- Designed a context-aware LLM layer with dynamic prompt engineering and real-time context injection, reducing repeated-query resolution time by 40–45% and cutting manual faculty intervention through student self-service access.
- Improved vector retrieval performance by 35% by systematically tuning FAISS index parameters and adding a Redis caching layer that held low latency across concurrent sessions during peak academic demand.

Programmer Analyst — Machine Learning
Cognizant Technology Solutions | India

Aug 2021 – Nov 2022

- Deployed 3 automated batch pipelines on AWS processing daily incentive compensation files for a U.S. medical device sales organization using Python, S3, Lambda, and SageMaker Batch Transform, cutting commission calculation cycle time by 40% and guaranteeing county-level payout calculations within every 24-hour window.
- Eliminated 55% of manual business-rules processing by engineering an NLTK document parsing pipeline that automatically extracted compensation rules, territory hierarchies, and manager reassignments from daily flat-file feeds, removing error-prone human interpretation across continuous operations.
- Increased incentive-payout forecasting accuracy by 35% by deploying XGBoost classification models on AWS SageMaker trained on multi-year sales performance data to predict commission attainment tiers across U.S. county-level territories, retiring legacy spreadsheet-based calculations entirely.
- Owned CI/CD across a cross-functional ML team and mentored junior engineers, standardizing AWS CodePipeline, Git branching, and CloudWatch alerting to achieve zero-downtime deployments across all 3 production incentive pipelines running 24/7.

Machine Learning Engineer
Lemoius Pvt. Ltd. | India

May 2020 – Aug 2021

- Pioneered a real-time content-based job recommendation engine for a 20,000+ profile marketplace using TF-IDF vectorization and cosine-similarity ranking over candidate profiles and job description text, lifting candidate-to-role match relevance by 35% and replacing manual recruiter shortlisting.
- Reduced candidate profile processing time by 30% by engineering an end-to-end NLP preprocessing pipeline with NLTK covering tokenization, stopword removal, lemmatization, and n-gram extraction to turn raw candidate data into clean, model-ready feature vectors at scale.
- Accelerated recruiter shortlisting efficiency by 25% by productionizing a real-time top-N candidate ranking system with scikit-learn TF-IDF and cosine-similarity scoring, delivering ranked candidate lists for every new job posting with zero manual intervention.
- Cut model iteration cycles by 40% by standardizing reproducible experiments with scikit-learn pipelines and Git version control, enabling the team to evaluate, validate, and ship updated matching models within a single sprint.

CERTIFICATIONS

Microsoft Certified: Azure AI Engineer Associate (AI-102)
AWS Certified AI Practitioner – Foundational

Microsoft
Amazon Web Services

EDUCATION

M.S., Data Science — Concentration: Applied NLP & Generative AI
University of North Texas | Denton, TX

B.Tech., Computer Science & Engineering — Major: Machine Learning & Deep Learning
Kalasalingam University | India